

Recursive Method

Definition:

A **recursive method** is a method in which a function **calls itself** repeatedly until a **base condition** is reached.

Algorithm

1. Start.
2. Define the base condition.
3. If the base condition is true, return the result.
4. Otherwise, call the same function again with a smaller/simpler input.
5. Repeat until the base condition is reached.
6. Return the result.
7. End.

Output:

```
Enter a number: 5
Factorial = 120

...Program finished with exit code 0
Press ENTER to exit console.
```

Time Complexity

It depends on the number of recursive calls.

- One recursive call for each input → **$O(n)$**
- Two recursive calls at each step → often **$O(2^n)$**

Space Complexity

Recursive methods require memory for the **function call stack**.

- Recursion depth of $n \rightarrow O(n)$ space.